Corrected and conduct of chapter 6

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. 6. *Action:* Resuming rate 2 $a+h$.
2. $\mathbb{R}\setminus\{2\}$. *Action:* Cancel denominator.
3. Positive outside of [-2; 1], negative between roots, zero at bounds. *Action:* Build two lines of signs.
4. x^3-2x^2+x-2. *Action:* Distribute each term without omission.
5. They are equal. *Action:* Replace -x in parentheses.
6. $y=2$. *Action:* Resuming the tangent equation and simplifying.

Research activity

x^3 is strictly increasing with horizontal tangent in 0. For an increasing function derivatable, the rates on both sides are positive or zero, so the derivative cannot be negative.

Immediate applications

1. $5x^4$ on $\mathbb{R}$; $-\dfrac{2}{x^3}$ for $x\neq 0$; $\dfrac{1}{2\sqrt{x}}$ for $x>0$.
2. $2x(3x-2)+3(x^2+1)=9x^2-4x+3$; $-\dfrac{2x}{(x^2+1)^2}$ on $\mathbb{R}$.
3. $f\prime=2x-4$, so decrease to 2 then increase. Minimum -3 in 2 ; values at bounds 1 and -2, so maximum 1 in 0.
4. First pair, second odd; difference $x^2+1-2x=(x-1)^2\geq 0$, equal to 1.
5. $A(x)=x(6-x)$ on [0; 6] ; $A\prime=6-2x$. Maximum 9 m ^{2} in 3.

Exercises: approach and results

Exercise 6.1 · A collection of derivatives. $12x^3-2$; $2x(x-1)+(x^2+2)=3x^2-2x+2$; $\dfrac{[2(x+3)-(2x+1)]}{(x+3)^2}=\dfrac{5}{(x+3)^2}$ for $x\neq -3$.

Exercise 6.2 · A complete picture. $f\prime=3(x-1)(x-3)$; signs + , −, +. Values $f(0)=1$, $f(1)=5$, $f(3)=1$, $f(4)=5$. Minimum 1 in 0 and 3; maximum 5 in 1 and 4.

Exercise 6.3 · A quotient. $f\prime=1-\dfrac{4}{x^2}=\dfrac{(x-2)(x+2)}{x^2}$. Negative sign before 2, positive after. Minimum 4 in 2 ; for $x>0$, $x+\dfrac{4}{x}\geq 4$, equality only in 2.

Exercise 6.4 · Compare two curves. $f-g=x(x-1)(x+1$. Signs: negative on] $-\infty;-1[$, positive about] $-1;0[$, negative on]0; 1[, positive on]1; + $\infty[$. Intersection in $-1,0,1$. One can do the table of three factors without derivation.

Exercise 6.5 · One package. $V=x(12-2x)^2=4x(6-x)^2$. $V'=12(x-2)(x-6)$, positive on [0; 2[, negative on]2; 6[. $V(0)=V(6)=0$, $V(2)$ = 128 cm ^{3}. Maximum for x=2 cm; box 8 $\times 8\times 2$.

Exercise 6.6 · Stationary without extremum. $f'=3x^2$, so value 0 in 0. In any vicinity of 0, negative x give $f(x)<0$ and positive x $f(x)>0$; no extremum.

Exercise 6.7 · Compare strategies. Square: $(x-3)^2+3\geq 3$. Derivative: $2x-6$, minimum in 3, value 3. The first avoids a complete study of variation; the second illustrates the link between Chapters.

Fiscal year 6.8 · Deferred review. It decreases on] $-\infty;0[$ and]0; $+\infty[$. It does not decrease on their meetings: $-1<1$ but $f(-1)=-1<1=f(1)$. The domain is not an interval.

CQFD, Exercise 12 p. 156, Questions 1 and 2. $f(x)-g(x)=\frac{1}{2}x^2-1=\frac{1}{2}(x-\sqrt{2})(x+\sqrt{2})$. The f curve is strictly above that of g for $x<-\sqrt{2}$ or $x>\sqrt{2}$, below between these two numbers; equality to the apses $\pm\ \overline{2}$.

CQFD, exercise 16 p. 156, questions f and g. For $f(x)=x^3$, we can choose] $-\infty;0[$: $f(x)<0$ and the function is strictly increasing. $g(x)=x^2-4$, one can choose [0; 2[: $g(x)<0$ and the function is strictly increasing over this interval. Sub-intervals not reduced to one point Also suitable.

MCQ: answers and reasons

MCQ 1.

1. B. Whole power.

2. A. Inverse rate.

3. D. Product rule.

4. C. Sign of derivative and variation.

5. B. Non-zero denominator.

MCQ 2.

1. C. $f(-x)=-f(x)$.

2. A. The other conclusions call for more.

3. D. Formula and field.

4. B. Achieved in 1.

5. C. Derivative $2x_0=0$.

True or false

1. False: $f(x) = \dfrac{1}{x} > 0$ on]0; + ∞[, but $f\prime(x) < 0$.

2. True, with the assumptions specified; at the bounds, the conclusion is not imposed.

3. False without sign assumptions: on [$-2; -1]$, $u(x) = v(x) = x$ growth, but x^2 Decline.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.